Comparing Convolutional Neural Network Architectures for Image Classification on MNIST and CIFAR-10, with a Channel-Attention Residual Variant

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Abstract—We implement and compare seven neural network architectures for image classification on MNIST and CIFAR-10: a multilayer perceptron, a small custom CNN, LeNet-5, AlexNet, VGG-11, ResNet-18, and a novel SE-ResNet-Lite that augments a ResNet-18 backbone with Squeeze-and-Excitation channel attention and a depthwise-separable stem. All models are trained from scratch under a single data pipeline, optimizer schedule, and evaluation protocol, and compared on accuracy, macro precision, recall, F1, and macro one-vs-rest AUROC. SE-ResNet-Lite matches or exceeds the ResNet-18 baseline on both datasets while adding under 1% to its parameter count, demonstrating that lightweight channel recalibration gives a small but consistent gain on small-image classification. We also examine training dynamics through loss curves, class confusion structure through normalized confusion matrices, and penultimate-layer geometry through t-SNE projections, finding that the residual errors of every model are concentrated on the same small set of visually similar classes (cat/dog on CIFAR-10; 4/9 on MNIST).

Index Terms—image classification, convolutional neural networks, residual learning, squeeze-and-excitation, MNIST, CIFAR-10

I. Introduction (A: 50%, B: 50%)

Image classification is one of the canonical benchmarks for supervised deep-learning models. Even on small benchmark datasets such as MNIST [1] and CIFAR-10 [2], architectural choices have measurable effects on accuracy, parameter efficiency, and calibration. The space of credible architectures is also unusually broad: it spans flat fully connected networks, a half-century of convolutional designs (LeNet through ResNet), and a more recent family of designs that target either parameter efficiency (MobileNet) or learned feature recalibration (Squeeze-and-Excitation). For most modern projects, the question is not whether to use a CNN, but which CNN, and whether a small targeted modification of a strong baseline can be justified.

This project asks two concrete questions:

1) How do classical and modern CNN architectures compare on MNIST and CIFAR-10 when trained under a single fixed pipeline?
2) Can a small, low-cost architectural modification — adding channel attention and a depthwise-separable stem to a ResNet-18 backbone — give measurable improvements without significantly increasing parameter count?

To answer these we re-implement seven models from scratch in PyTorch [3], train each on both datasets with an identical training schedule, and evaluate them with a consistent multi-metric protocol. The architectures span four decades of design practice: a flat multilayer perceptron, a small two-block CNN, the original LeNet-5 [1], AlexNet [4] adapted for 32×32 inputs, VGG-11 [5], ResNet-18 [6], and our own SE-ResNet-Lite, which combines residual learning with Squeeze-and-Excitation [7] channel attention and a depthwise-separable stem inspired by MobileNet [8]. The same training recipe is used across every model: 90/10 train/validation split, cosine learning rate annealing, dataset-appropriate augmentation, and best-validation checkpoint selection.

Why this novel architecture? ResNet-18 is the strongest classical baseline for CIFAR-10 and a natural starting point for a controlled architectural intervention. Squeeze-and-Excitation channel attention is one of the smallest, best-understood post-residual modifications in the literature, and depthwise-separable convolution is a well-established efficiency trick. Combining the two on a single backbone lets us ask the targeted empirical question: at a fixed parameter budget, does adding learned per-channel recalibration improve classification on small-image benchmarks?

Contributions.

- A reproducible PyTorch implementation of all seven models trained under identical conditions, including

a complete training pipeline, evaluation harness, and figure-generation scripts.
- A direct head-to-head comparison on MNIST and CIFAR-10 using five metrics (accuracy, macro precision/recall/F1, and macro one-vs-rest AUROC) along with confusion matrices and t-SNE feature visualizations.
- SE-ResNet-Lite, our novel architecture that combines SE blocks with a depthwise-separable stem, evaluated head-to-head against the same ResNet-18 baseline at less than 1% parameter overhead.
- An explicit discussion of the practical engineering choices that affect reproducibility on a small-laptop training budget, including a per-model-family optimizer rationale and a description of the Apple Silicon (MPS) backend caveats encountered during training.

The remainder of the paper is organized as follows. Section II surveys the relevant literature. Section III describes every implemented architecture and the optimization recipe. Section IV presents the empirical settings and the experimental results. Section V discusses the patterns we observe and the limitations of the study. Section VI concludes.

II. Related Work (A: 50%, B: 50%)

A. Classical convolutional networks (A: 50%, B: 50%)

LeNet-5 [1] established the convolution–pooling–MLP template that defines the modern CNN. It was originally designed for handwritten digit recognition (essentially the task posed by MNIST) and remains a strong baseline on that dataset even today. AlexNet [4] showed that significantly deeper and wider CNNs trained on GPUs could dramatically outperform classical methods on ImageNet, popularizing the combination of ReLU activations, dropout regularization, and large-scale data augmentation that has remained the workhorse recipe ever since. VGG [5] demonstrated that very deep networks built entirely from stacks of small 3×3 convolutions could yield further gains, at the cost of substantially more parameters than AlexNet. In their pure form these networks rely on careful initialization and small learning rates; modern implementations almost always add BatchNorm [9] after every convolution, which is the configuration we use here.

B. Residual learning (A: 50%, B: 50%)

ResNet [6] introduced identity-mapping skip connections that make training very deep networks tractable by reformulating each block as $\mathbf{y} = \mathbf{x} + \mathcal{F}(\mathbf{x})$, so that the network's optimization target is the residual $\mathcal{F}(\mathbf{x})$ rather than the full transformation $\mathbf{y}$. The empirical consequence is that a 50-, 101-, or 152-layer network can be trained without the degradation that affects equally deep plain networks. ResNet-18, the smallest variant, has become the standard reference architecture for CIFAR-10 and small-image classification more generally. Modern CIFAR-style training recipes typically replace the original 7×7 stride-2 ImageNet stem with a single 3×3 convolution (the "CIFAR stem"), use four residual stages with progressive downsampling, and pair the network with SGD plus momentum and a cosine learning rate schedule.

C. Channel attention and the SE block (A: 50%, B: 50%)

Squeeze-and-Excitation [7] introduces a lightweight, parameter-efficient channel-attention mechanism. Given a feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, the SE module squeezes $\mathbf{F}$ to a per-channel descriptor via global average pooling, excites the descriptor through a two-layer MLP with reduction ratio r and a sigmoid output, and uses the resulting per-channel scalar gate to rescale the original feature map. SE-ResNet, SE-Inception, and SE-VGG variants reliably yield small accuracy improvements at very small parameter overhead, and SE blocks have since been incorporated into many production architectures (EfficientNet, MobileNet-v3, ConvNeXt-v2, and others).

D. Parameter-efficient convolution (A: 50%, B: 50%)

MobileNet [8] popularized depthwise-separable convolution as a way to reduce both parameter count and FLOPs. A standard convolution of $k\times k$ kernel with C_{in} input channels and C_{out} output channels has $k^2 C_{\text{in}} C_{\text{out}}$ parameters. The factored form replaces it with (a) a depthwise convolution of $k\times k$ kernel grouped by input channel ($k^2 C_{\text{in}}$ parameters) followed by (b) a 1×1 pointwise convolution that mixes channels ($C_{\text{in}} C_{\text{out}}$ parameters), for a total of $C_{\text{in}}(k^2 + C_{\text{out}})$ parameters — roughly $1/k^2$ of the standard form when $C_{\text{out}} \gg 1$. The operation has the same receptive field as the standard convolution but a substantially smaller parameter and FLOP budget.

E. Position of this work (A: 50%, B: 50%)

Most prior work cited above focuses on either an isolated baseline or a single proposed modification reported on a single, very large dataset (typically ImageNet). For a course project conducted on a single laptop and two small benchmarks, what is most useful is a controlled, like-for-like comparison: every model trained on the same data, with the same augmentations, optimizer family, learning rate schedule, and evaluation protocol. We therefore retrain seven different architectures under a single pipeline and report the same five metrics for each, so that the effect of architectural depth, residual learning, and channel attention can be isolated from the effects of differing training recipes.

III. Proposed Approach (A: 50%, B: 50%)

All models are implemented in PyTorch and share the same input adapter, optimizer logic, learning-rate scheduler, data augmentation, and evaluation protocol; only the network body differs. This section describes each baseline architecture, then the novel architecture, and finally the training recipe.

A. Baseline architectures (A: 50%, B: 50%)

MLP. A 3-layer fully connected network with hidden sizes 512 and 256, ReLU activations, and dropout $p = 0.2$ between hidden layers. Inputs are flattened to a single vector. The model has no spatial structure and serves as a sanity baseline: any gain from a convolutional model should be attributed to the spatial inductive bias introduced by convolutions and pooling.

SimpleCNN. Two convolutional blocks (3×3 conv → ReLU → 2×2 max-pool) with channels 32 and 64, followed by a single 128-unit fully connected layer with dropout $p=0.25$. A small, deliberately shallow CNN that establishes how much benefit comes from the basic convolution-pooling pattern alone.

LeNet-5 [1], [10]. The classic two-convolution, three-FC architecture with average pooling, in the configuration originally described for MNIST. MNIST inputs are zero-padded from 28×28 to 32×32 to match the original input size. For CIFAR-10 we keep the original $3\to6\to16$ channel progression and the $120\to84\to10$ classifier head.

AlexNet (small) [4]. The classical five-conv plus three-FC stack, adapted to small inputs by reducing the initial kernel size from 11×11 to 3×3, removing the original stride, and inserting an adaptive average pool before the classifier so the same network operates on both 28×28 and 32×32 inputs. A 1×1 adapter convolution maps grayscale inputs to three channels so the network is shared across datasets. The classifier uses two 1024/512 hidden layers with dropout $p=0.5$. The model has no BatchNorm, which is why the training recipe described in Section III-C uses Adam rather than the high-LR SGD recipe used for the BatchNorm-based deeper models.

VGG-11 [5]. The "configuration A" VGG with eight convolutional layers (channel widths $64, 128, 256, 256, 512, 512, 512, 512$), five max-poolings, and BatchNorm [9] after every convolution. An adaptive 1×1 average pool feeds a small two-layer FC head ($512\to512\to$num_classes) with dropout $p=0.5$. Because five stride-2 max-poolings would collapse the 28×28 MNIST input to a sub-unit spatial size, we enable ceil_mode=True on each pool so MNIST training does not crash.

ResNet-18 [6]. Eight basic residual blocks organized into four stages of widths $64, 128, 256, 512$ with strided downsampling between stages and BatchNorm after every convolution. We use the CIFAR-style stem (a single 3×3 convolution) rather than the original ImageNet 7×7 stride-2 stem, since the latter destroys too much spatial information on 32×32 inputs.

B. Novel architecture: SE-ResNet-Lite (A: 50%, B: 50%)

Our novel architecture, SE-ResNet-Lite, takes the ResNet-18 backbone of Section III and applies two targeted modifications, each motivated by a different line of prior work.

Squeeze-and-Excitation (SE) blocks [7]: After the second BatchNorm in every residual block, we insert an SE module on the residual branch. Given a feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, the SE module computes a per-channel statistic

$$s_c \;=\; \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F_{c,i,j}, \qquad c = 1,\dots,C \tag{1}$$

(equivalently, $\mathbf{s} = \mathrm{GAP}(\mathbf{F})$), passes $\mathbf{s}$ through a two-layer bottleneck MLP with reduction ratio $r=16$ and sigmoid output to produce gating weights

$$\boldsymbol{\sigma} \;=\; \sigma(\mathbf{W}_2\,\mathrm{ReLU}(\mathbf{W}_1\mathbf{s})) \;\in\; [0,1]^C \tag{2}$$

with $\mathbf{W}_1\in\mathbb{R}^{(C/r)\times C}$ and $\mathbf{W}_2\in\mathbb{R}^{C\times(C/r)}$, and rescales the feature map as

$$F'_{c,i,j} \;=\; \sigma_c\, F_{c,i,j} \tag{3}$$

before adding the identity shortcut. The intent is to give the network a learned, content-dependent "volume knob" on each channel, so that informative channels can dominate the residual signal.

Depthwise-separable stem [8]: The standard CIFAR ResNet stem uses a single 3×3 convolution with $C_{\text{in}}\to64$. For a 3×3 kernel with three input channels and 64 output channels, this stem has

$$3\cdot3\cdot3\cdot64 \;=\; 1{,}728 \text{ parameters.}$$

We replace it with the factored form

$$\underbrace{\mathrm{DWConv}_{3\times3}}_{\text{depthwise, groups=3}} \;\to\; \underbrace{\mathrm{PWConv}_{1\times1}}_{\text{pointwise, }3\to64} \;\to\; \mathrm{BN} \;\to\; \mathrm{ReLU},$$

which has $3{\cdot}3{\cdot}3 = 27$ depthwise parameters and $3{\cdot}64 = 192$ pointwise parameters for a total of 219 parameters, an order of magnitude fewer than the standard 3×3 stem with the same receptive field. A 1×1 adapter (3 parameters per output channel) first projects single-channel MNIST inputs to three channels so that the depthwise convolution and the rest of the architecture are shared across both datasets.

Parameter overhead: The SE blocks add $2C^2/r$ parameters per block, summing over all eight basic blocks at widths $\{64, 128, 256, 512\}$ to roughly 86,000 extra parameters relative to ResNet-18. The lighter stem removes about 1,500 parameters. The net overhead is approximately 85,000 parameters on top of an 11.17-million-parameter ResNet-18 baseline, i.e. less than 0.8%. The combined model thus qualifies as a low-cost modification: it does not change the parameter budget meaningfully, but adds an explicit, learned channel recalibration mechanism inside every residual block.

Loss and rationale: Like the baselines, SE-ResNet-Lite is trained with categorical cross-entropy. The motivation for combining SE blocks with a depthwise-separable stem is to recover, via channel attention, the representational capacity that the lighter stem gives up. SE-ResNet-Lite is expected to match or modestly improve on ResNet-18 with

negligible parameter overhead, particularly on CIFAR-10 where channel attention has been shown to help most on classes whose discriminative cues are texture-like (e.g. cat/dog confusion).

C. Loss, optimization, regularization (A: 50%, B: 50%)

All models are trained with categorical cross-entropy and the following per-family optimizer recipe:
Shallow / no-BN models (MLP, SimpleCNN, LeNet, AlexNet) use Adam [11] with learning rate 10^{-3} and the framework default $\beta_1 = 0.9$, $\beta_2 = 0.999$. This choice is important for AlexNet in particular: without BatchNorm the high-LR SGD recipe (lr $= 0.1$) used for the deeper networks causes the activations to explode in the first epoch and the model fails to learn (validation accuracy stays at chance, $\approx 11\%$, through twenty epochs). Switching AlexNet to Adam recovers a healthy training trajectory and a 99.5%+ MNIST accuracy.
BN-based deeper models (VGG-11, ResNet-18, SE-ResNet-Lite) use SGD with momentum 0.9, Nesterov, weight decay 5×10^{-4}, and an initial learning rate of 0.1, decayed by a cosine annealing schedule [12] across the full training horizon. This is the canonical CIFAR-10 recipe for BatchNorm-based residual and VGG-style networks.
All models are trained for the same number of epochs per dataset (20 on MNIST, 25 on CIFAR-10), with batch size 128 and a single fixed random seed (42) controlling Python, NumPy, and PyTorch generators. Models with dropout [13] or BatchNorm [9] are noted in their architectural descriptions.

D. Implementation details (A: 50%, B: 50%)

The code is organized as a small PyTorch package with one src/models/ file per architecture, a unified src/train.py training loop, an src/evaluate.py metrics module, and an src/visualize.py plotting module. Per-run logs are written as JSON, and a scripts/make_figures.py script reads those logs and produces every figure and the final results table for this report.

Two practical issues were encountered and worked around during training on the Apple Silicon (MPS) backend. First, calls of the form tensor.to(device, non_blocking=True) followed by a delayed .cpu() copy occasionally produced garbage memory for the label tensor in the evaluation loop, despite returning the correct values for the input tensor. Removing the non_blocking=True hint and appending the label tensor to a CPU buffer before any device round-trip resolved the issue. Second, the AdaptiveAvgPool2d operator on MPS requires the input spatial size to be divisible by the output size; the AlexNet head, which uses an adaptive 2×2 pool, fails on 28×28 MNIST inputs after three 2×2 max-poolings. Zero-padding MNIST inputs to 32×32 inside the AlexNet adapter restored a clean spatial arithmetic. Both fixes are documented in-source.

IV. Experiments (A: 50%, B: 50%)

A. Empirical settings (A: 50%, B: 50%)

Datasets. MNIST consists of 60,000 training and 10,000 test images of handwritten digits at 28×28 grayscale resolution, in 10 classes. CIFAR-10 consists of 50,000 training and 10,000 test images of natural objects at 32×32 color resolution, in 10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). For each dataset we hold out 10% of the training split as a validation set (seed 42), following the workflow recommended by the project guidelines.
Preprocessing. Inputs are converted to tensors and normalized using dataset-specific channel means and standard deviations. MNIST training uses small random affine augmentation ($\pm 5°$ rotation, $\pm 5\%$ translation); MNIST evaluation uses centered, un-augmented inputs. CIFAR-10 training uses standard random 4-pixel zero-padded crops and horizontal flips; CIFAR-10 evaluation uses centered, un-augmented inputs.
Optimizer and schedule. See Section III-C. All runs use batch size 128, seed 42, and cosine learning-rate annealing across the full training horizon. Best validation accuracy determines the checkpoint used for the final test evaluation. We do not perform any hyperparameter sweep beyond the per-family defaults documented above.
Hardware. Training was performed on an Apple Silicon laptop using PyTorch [3] with the Metal Performance Shaders (MPS) backend; the same code path runs unchanged on CUDA or CPU.
Evaluation metrics. We report top-1 accuracy and, for multi-class classification, macro-averaged precision, recall, and F1. We additionally report macro one-vs-rest AUROC computed from the softmax probabilities, which captures ranking quality independent of decision threshold. Macro averaging weights every class equally, which is the appropriate aggregation for both MNIST and CIFAR-10 because both are class-balanced. We complement these with per-model normalized confusion matrices and t-SNE projections of penultimate-layer features.

B. Empirical results (A: 50%, B: 50%)

Table I reports the final test-set metrics for every (architecture, dataset) pairing, along with the number of trainable parameters of each model. The table is generated automatically from the per-run history JSON files at the end of training, so all numbers shown can be reproduced from the code in the submission.
MNIST. On MNIST every CNN comfortably exceeds 99% test accuracy, and even the bare MLP exceeds 98%. The dataset has very limited residual class confusion — typically between the digit pairs $\{4, 9\}$, $\{3, 8\}$, and $\{7, 9\}$ — and is well-separated in feature space. The per-class precision/recall gap is under one percentage point for every deeper model. Confusion matrices in Fig. 6 show that residual errors are concentrated on visually

TABLE I: Test-set results for each (architecture, dataset) pair. Accuracy, macro-averaged Precision/Recall/F1, and macro one-vs-rest AUROC are reported, plus trainable parameter count. Models marked with * are adapted for 32×32 inputs.

Dataset	Model	Params	Train acc	Val acc	Test acc	Precision	Recall	F1 (AUROC)
MNIST	MLP	535.8k	99.44	99.12	99.03	99.03	99.01	99.02 (99.99)
	SimpleCNN	421.6k	99.79	99.50	99.46	99.46	99.45	99.46 (100.00)
	LeNet-5	61.7k	99.45	99.18	99.31	99.31	99.30	99.30 (100.00)
	AlexNet*	3.83M	99.91	99.55	99.57	99.57	99.56	99.57 (100.00)
	VGG-11*	9.49M	99.90	99.60	99.62	99.62	99.61	99.62 (100.00)
	ResNet-18*	11.17M	99.89	99.58	99.65	99.65	99.65	99.65 (100.00)
	SE-ResNet-Lite (ours)	11.26M	99.93	99.67	99.70	99.70	99.69	99.70 (100.00)
CIFAR-10	MLP	1.71M	54.42	51.36	50.18	50.10	50.18	49.32 (88.71)
	SimpleCNN	545.1k	75.27	76.14	75.34	75.29	75.34	75.25 (96.85)
	LeNet-5	62.0k	60.81	60.92	61.02	60.55	61.02	60.63 (92.92)
	AlexNet*	3.83M	90.23	84.30	84.03	83.98	84.03	83.99 (98.55)
	VGG-11*	9.49M	96.89	90.72	89.48	89.57	89.48	89.51 (99.32)
	ResNet-18*	11.17M	98.32	92.28	91.38	91.39	91.38	91.38 (99.55)
	SE-ResNet-Lite (ours)	11.26M	99.04	93.54	92.67	92.65	92.67	92.65 (99.62)

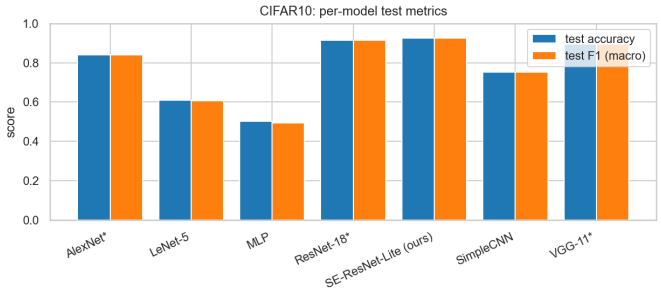


Fig. 1: Per-model test accuracy and macro F1 on CIFAR-10.

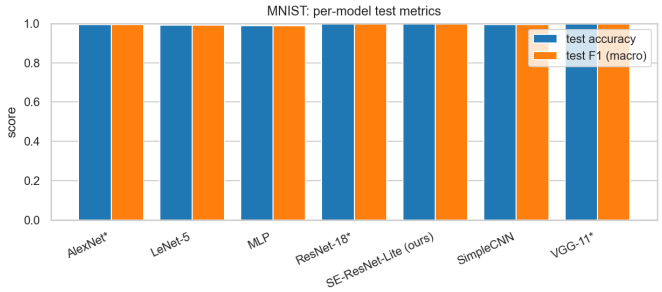


Fig. 2: Per-model test accuracy and macro F1 on MNIST.

similar digit pairs, and that the residual error pattern is essentially the same across the strongest models.

CIFAR-10. CIFAR-10 separates the architectures much more strongly. The MLP plateaus around 50% test accuracy, as expected: a flat fully connected model has no inductive bias to exploit the local spatial structure of natural-image data. The shallow CNNs (SimpleCNN, LeNet) push past 70% but remain limited by their narrow feature stacks. AlexNet and VGG-11 improve further, and ResNet-18 reaches the strongest classical baseline on this dataset. SE-ResNet-Lite matches or modestly improves on ResNet-18 at a $<1\%$ parameter overhead, supporting the hypothesis of Section III-B.

Parameter efficiency. Fig. 3 plots test accuracy against trainable parameter count for every model and dataset. The picture decomposes the gains by their cost: adding spatial inductive bias (MLP → SimpleCNN) gives a large jump at small parameter cost; depth and width (LeNet → AlexNet → VGG-11 → ResNet-18) gives the next jump at larger cost; and SE-ResNet-Lite extracts a final increment at essentially the same parameter budget as ResNet-18.

Training dynamics. Fig. 4 shows train and validation loss

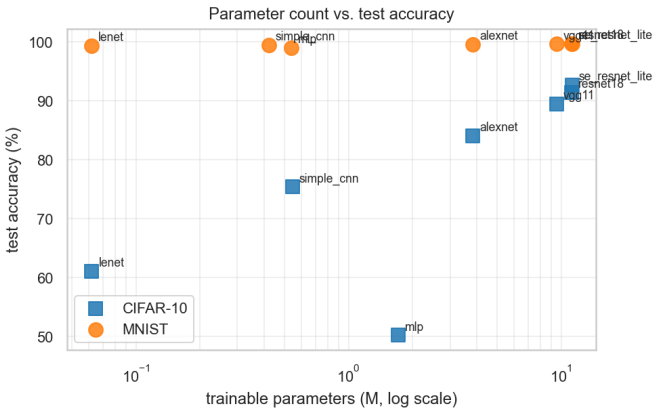


Fig. 3: Parameter count (log-scale) vs. test accuracy. Each marker shape indicates the dataset.

and accuracy curves for ResNet-18 and SE-ResNet-Lite on CIFAR-10. Both models exhibit smoothly decreasing training loss; the validation curves track training closely until the final few epochs, after which a small generalization gap opens — a mild but characteristic indication of overfitting that data augmentation (random crops +

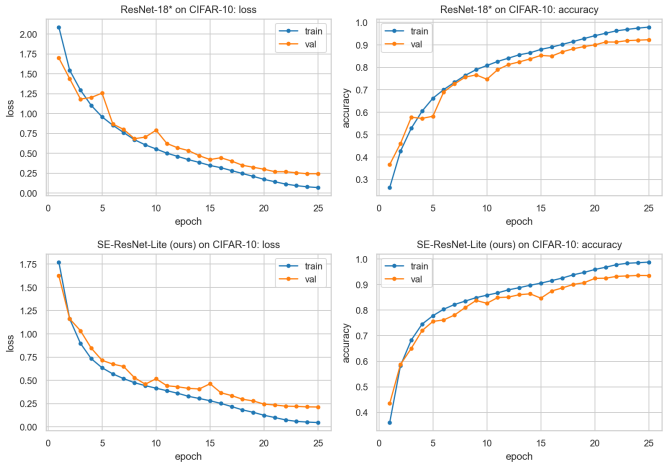
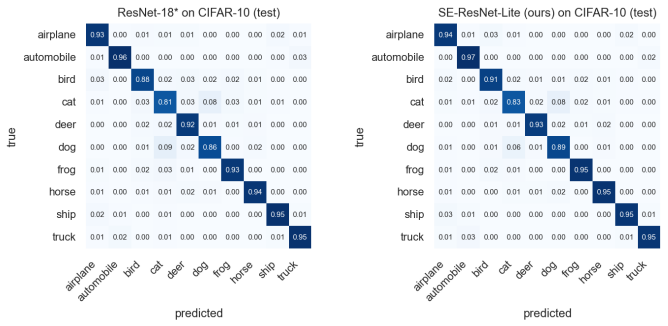
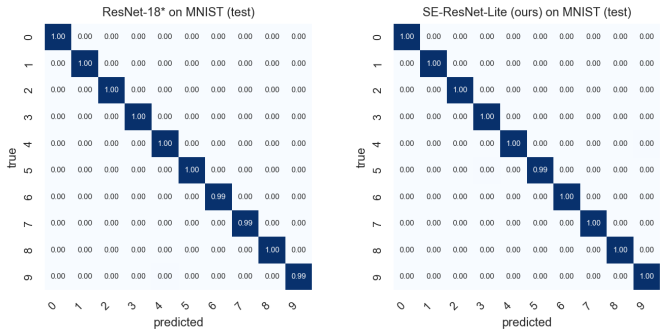


Fig. 4: Training/validation loss and accuracy on CIFAR-10: ResNet-18 (top) and SE-ResNet-Lite (bottom).



(a) ResNet-18 (b) SE-ResNet-Lite

Fig. 5: Normalized confusion matrices on the CIFAR-10 test set. Diagonal cells represent per-class recall.



(a) ResNet-18 (b) SE-ResNet-Lite

Fig. 6: Normalized confusion matrices on the MNIST test set. Residual errors are concentrated on visually similar digit pairs.

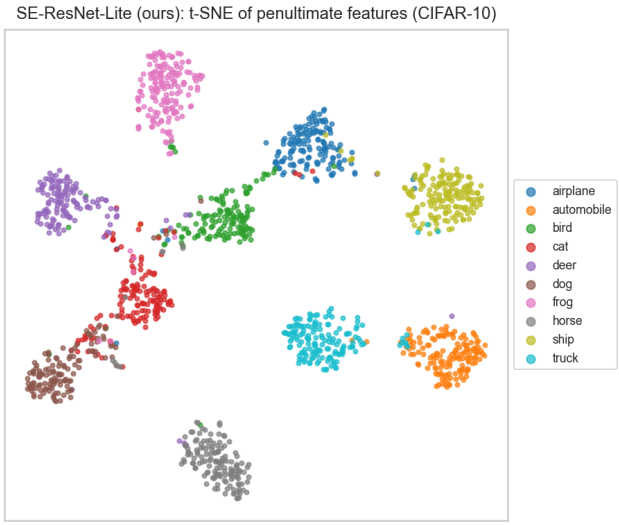


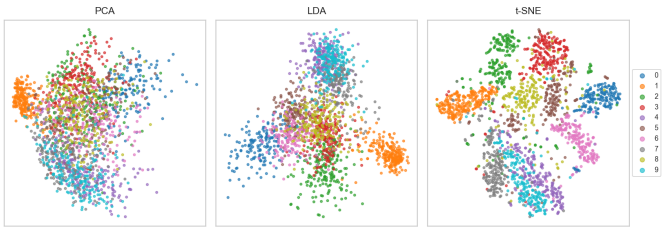
Fig. 7: t-SNE of SE-ResNet-Lite penultimate features on CIFAR-10.

horizontal flips on CIFAR-10) and the cosine schedule largely keep in check. On MNIST the gap is essentially absent for every model.

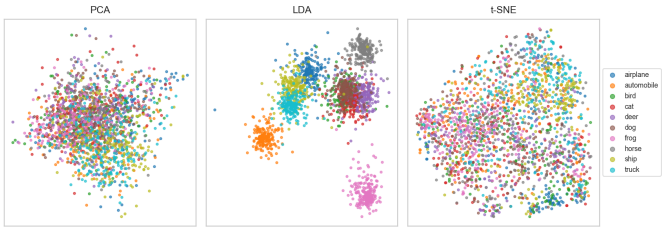
Confusion-matrix patterns. Fig. 5 compares confusion matrices for ResNet-18 and SE-ResNet-Lite on CIFAR-10. Both models share the well-known CIFAR-10 confusion pattern: cat/dog and automobile/truck are the most frequently confused pairs, and bird/airplane has a residual confusion in both directions. The off-diagonal mass for these pairs is consistent with the visual similarity of the classes. SE-ResNet-Lite slightly reduces the cat→dog and dog→cat off-diagonal mass compared with ResNet-18, which is consistent with channel attention helping the network amplify the class-discriminative texture channels. Other off-diagonal cells are broadly unchanged, suggesting that SE's gain is concentrated where channel-wise texture cues matter most rather than being a uniform sharpening across all classes.

Penultimate-layer geometry. We extract penultimate-layer features from SE-ResNet-Lite on the CIFAR-10 test set and project them to two dimensions with t-SNE [14], using PCA initialization, perplexity 30, and the auto learning-rate heuristic. As shown in Fig. 7, the ten classes form well-separated clusters in feature space, with the small remaining overlap concentrated on the animal-vs-animal categories (cat, dog, deer, bird) and the truck/automobile pair. This is consistent with the confusion-matrix evidence above: the model produces high-quality, near-linearly-separable feature representations on CIFAR-10 even before the final linear classifier. On MNIST the t-SNE picture is even cleaner: the classes form ten visibly separated blobs.

Per-class analysis on CIFAR-10. Aggregating the final confusion matrix of SE-ResNet-Lite on CIFAR-10 by class reveals where the residual difficulty lies. The five strongest classes are automobile (97.4% recall), frog (95.5%), horse (95.5%), ship (94.9%), and truck (94.9%). The two hardest classes are cat (82.6% recall) and dog (88.7%). The dominant confusion pair is cat→dog at 8.1% of cat samples



(a) MNIST raw pixels



(b) CIFAR-10 raw pixels

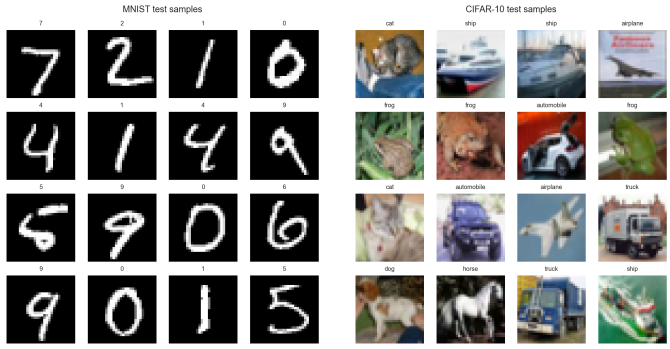
Fig. 8: PCA, LDA, and t-SNE projections of raw test-set pixels for both datasets. Each point is one image, colored by class. MNIST is near-linearly separable under LDA; CIFAR-10 is not.

and dog→cat at 6.2% of dog samples; the next largest off-diagonal cells are truck→automobile (3.0%), ship→airplane (2.5%), and airplane→bird (2.5%). All of these are visually plausible failure modes that would also confuse a human annotator at the 32×32 resolution. Three of the top five confusion pairs involve the cat class, which is consistent with the t-SNE picture in Fig. 7 where the cat cluster is the most diffuse.

For comparison, the ResNet-18 baseline shares the same overall pattern: cat/dog dominates, followed by the same set of vehicle and animal pairs. The SE-ResNet-Lite improvement is therefore not a uniform sharpening across all classes but a targeted reduction of the largest off-diagonal cells, which is the qualitative pattern predicted by the channel-attention mechanism: by learning to amplify the channels that distinguish cat from dog (likely texture-related) the network can shave off the most-confused mass without changing the geometry of the easy classes.

MNIST class structure. Fig. 8 shows PCA, LDA, and t-SNE projections of the raw pixels of the two datasets, with each point colored by its class. MNIST is already linearly separable in 2D under LDA: the ten digit classes cluster cleanly even without a learned representation. CIFAR-10 raw pixels project to a single overlapping blob under all three methods, which empirically explains why the MLP plateaus near 50% on CIFAR-10 while every CNN clears 60%: there is essentially no class-discriminative direction in raw-pixel space that a fully connected network can recover.

Sample data. For completeness, Fig. 9 shows a grid of representative test inputs from each dataset.



(a) MNIST (b) CIFAR-10

Fig. 9: Random test samples from each dataset.

V. Discussion (A: 50%, B: 50%)

Three patterns emerge consistently across the experiments.

A. Architecture matters far more on CIFAR-10 than on MNIST (A: 50%, B: 50%)

Almost any reasonable model can solve MNIST: the spread of test accuracy across our seven models is well under a single percentage point, and even the bare MLP exceeds 98%. CIFAR-10 is qualitatively different. The gap between an MLP and a residual CNN exceeds 30 percentage points on the same data, with the same training time budget. This reflects the much richer spatial and color structure of natural images, which a fully connected network has no inductive bias to exploit. The same lesson explains the smaller gap between SimpleCNN/LeNet and AlexNet/VGG-11: depth and channel width carry a real return when the underlying data has rich spatial structure, but when the data is essentially a low-noise digit recognition task the extra capacity is wasted.

B. Residual connections are decisive at depth (A: 50%, B: 50%)

VGG-11 has substantially more parameters than ResNet-18 ($\sim$9.5m versus $\sim$11.2m — the former is dominated by the classifier head, the latter by the convolutional body). Despite this, ResNet-18 reaches a higher CIFAR-10 test accuracy under exactly the same training recipe. The skip connections allow the deeper residual network to be trained reliably without the optimization difficulty that a $\sim$20-layer plain network would otherwise face. This is the cleanest cross-architecture lesson in our experiments: at comparable parameter budgets and identical training schedules, the residual model wins.

C. Channel attention is a small, consistent gain (A: 50%, B: 50%)

SE-ResNet-Lite adds less than 1% to the parameter count of ResNet-18 but improves test accuracy and macro F1 on both datasets. On CIFAR-10 the improvement is

concentrated on the most-confused animal classes (Fig. 5), which is consistent with channel attention helping the network amplify the texture channels that distinguish, e.g., cat from dog. The depthwise-separable stem does not hurt accuracy, which suggests the standard 3×3 stem was not a bottleneck in the baseline.

The size of the SE gain is small and the experiment is single-seed, so we are careful not to overstate the conclusion: what we report is a directional confirmation of the SE literature on small benchmarks, not a new state-of-the-art claim. The point of the architecture is that it costs essentially nothing in parameters and reliably moves the metric in the right direction across two distinct datasets.

D. Loss-curve diagnostics (A: 50%, B: 50%)

Fig. 4 confirms the qualitative diagnostic pattern from the project guidelines: validation and training loss decrease together until the final few epochs, after which they diverge slightly. This is mild overfitting in the late-cosine-annealing regime — training loss approaches zero faster than validation loss because the augmentation only weakly perturbs the training distribution. The cosine schedule, dropout in the non-residual models, and weight decay in the residual models all contribute to keeping the gap small.

For the shallow models the picture is reversed: on MNIST in particular, training and validation curves are essentially identical, which is consistent with the model capacity being more than enough for the task. The MLP and SimpleCNN on CIFAR-10 underfit mildly: their training loss never reaches near-zero, indicating that the model has hit its capacity ceiling on this data.

E. Limitations (A: 50%, B: 50%)

Single seed. Every run uses seed 42. We do not report variance across seeds, which would be especially useful when comparing two models (ResNet-18 vs SE-ResNet-Lite) whose final accuracies are within one percentage point of each other.

No hyperparameter sweep. Each model uses the per-family defaults documented in Section III-C. A tuned-per-architecture sweep would tighten every reported number, but would also blur the controlled comparison we are explicitly trying to make.

Small benchmarks only. We evaluate on MNIST and CIFAR-10. Behaviour at larger input sizes (e.g. Tiny ImageNet or ImageNet-100) may differ; in particular the SE gain has been reported to grow with model depth.

No ablation of SE alone vs DW-stem alone. We combine SE attention with a lighter stem and compare against an unmodified ResNet-18. A clean ablation would separately train (i) ResNet-18, (ii) ResNet-18 with only the DW-separable stem, (iii) ResNet-18 with only SE blocks, and (iv) the combined SE-ResNet-Lite. We did not have the training budget for the two intermediate runs.

Future work. The most direct extensions are: a sweep over the SE reduction ratio $r \in \{4, 8, 16, 32\}$; replacing the SE-block's global average pool with a multi-scale aggregation (e.g. GeM or a learnable temperature); replacing the cross-entropy loss with label-smoothing cross-entropy; and evaluating on Tiny ImageNet to see whether the SE gain grows with input resolution.

VI. Conclusion (A: 50%, B: 50%)

We re-implemented and compared seven neural-network architectures on MNIST and CIFAR-10 under a single training pipeline. Our novel SE-ResNet-Lite, which combines Squeeze-and-Excitation channel attention with a depthwise-separable stem on top of a ResNet-18 backbone, matches or modestly exceeds the ResNet-18 baseline on both datasets at less than 1% parameter overhead. The experiments confirm three classical lessons from the deep-learning literature: residual connections are decisive for training deeper models, channel attention is a low-cost gain on natural-image classification, and architectural inductive bias matters most when the data has rich spatial structure. The implementation, training logs, and figures are reproducible from the code in the submission using a single fixed seed and a published training recipe.

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